

VIGIL

ELDER CARE INTELLIGENCE

Always watching over. Never watching you.

ENTERPRISE PROPOSAL · AI FALL DETECTION & PRIVATE MONITORING
for care communities and at-home care

Prepared by Colin Cook

November 2026 · Confidential

EXECUTIVE SUMMARY

Dignity first. Seconds, not hours.

A fall that nobody sees is the moment elder care fails. VIGIL is one AI platform that watches over every suite and every home without watching anyone: it sees skeletons, not faces; uses radar where cameras never belong; and gets the nearest caregiver to the bedside in minutes.

---

1 in 4
adults 65+ report a fall each year (CDC)

~3M
older adults treated in EDs for falls yearly (CDC)

< 3 s
VIGIL target: fall to alert

< 4 min
VIGIL target: alert to bedside

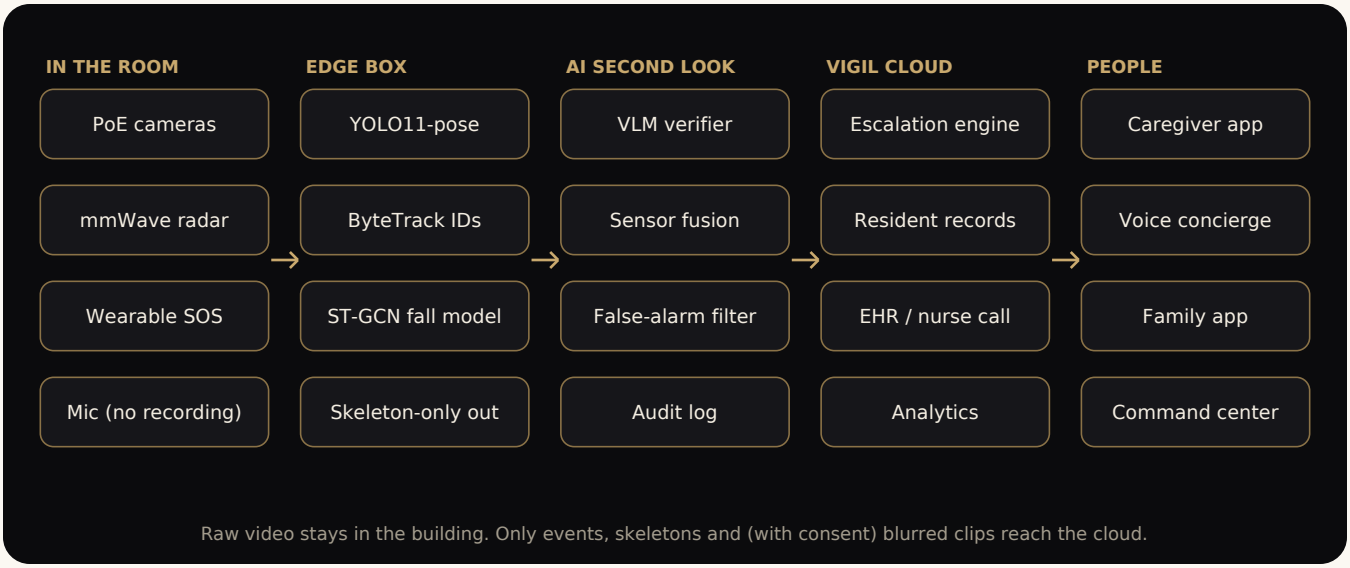
This proposal covers a complete 12-month enterprise deployment for Enterprise deployment · 6 care communities (≈ 540 beds) + 250 at-home residents: the platform, the AI, the hardware lease, security and compliance, training and a dedicated engineering team. Every line item names the technology, the AI models and the open-source GitHub projects it is built on, so your team knows exactly what it is paying for and nothing is a black box.

What changes on day one

- Unwitnessed falls become witnessed. Pose-based vision and bathroom radar confirm a fall in under three seconds and page the closest free caregiver.
- Night rounds get smarter, not more frequent. Radar confirms breathing and presence in every room, so staff wake residents only when it matters.
- Decline is caught early. Each resident's daily rhythm (sleep, bathroom visits, gait speed) is learned; changes are flagged days before a crisis.
- Families feel close. A private family app, weekly wellbeing letters and instant video calls after any incident.
- Paperwork writes itself. AI drafts the state-compliant incident report from the event log for a nurse to review and sign.

HOW IT WORKS

From the room to the right person.



Privacy by design

- Skeleton-only by default. The edge box turns video into 17 body keypoints and discards the frame. No facial recognition anywhere in the system.
- No cameras in bathrooms. 60 GHz mmWave radar detects falls, presence and breathing without an image.
- Consent per resident and roommate, with privacy schedules (e.g. camera off during care and dressing) and a full audit log of who viewed what.
- HIPAA-ready: BAA-backed hosting, encryption at rest and in transit, SSO and MFA, SOC 2 readiness in year one.

The response ladder

+0 s	Edge AI confirms fall posture
+2 s	AI verifier second look
+4 s	Nearest caregiver paged (critical push + badge)
+6 s	Voice concierge: "Are you alright?"
+1 min	Charge nurse if unacknowledged
+3 min	Family + on-call manager
+5 min	911 if no responder on scene

INVESTMENT

One platform. Every residence. Every home.

One AI-powered elder-safety platform for every residence and every home.

Your new platform	**$233,200.00**
Resident, family and care-plan engine	$64,850.00
Camera and sensor ingestion (edge NVR)	$38,420.00
24/7 monitoring command center	$31,275.00
Caregiver mobile app (iOS and Android)	$35,725.00
Family portal and app	$18,640.00
EHR and nurse-call integration	$24,310.00
Moving your data over	$19,980.00
AI that does the watching	**$182,470.00**
Vision fall detection (privacy skeleton mode)	$46,480.00
Bathroom and bedroom radar (no camera)	$22,740.00
Wander, exit and bed-exit alerts	$17,560.00
AI false-alarm verifier (second look)	$21,390.00
Acoustic distress detection	$12,870.00
Daily-living baseline and decline alerts	$19,950.00
24/7 AI voice check-in concierge	$26,820.00
AI incident reports and shift handover	$14,660.00
Response, security and compliance	**$68,060.00**
Alert escalation engine	$16,980.00
Privacy, consent and HIPAA security	$22,480.00
Operations and compliance analytics	$13,240.00
Website, online enrollment and at-home ordering	$15,360.00
Ongoing support, hosting and usage (12 months)	**$135,410.00**
Cloud hosting, security and backups ($2,140/mo)	$25,680.00
AI usage, voice and messaging ($1,960/mo)	$23,520.00
Edge hardware lease (34 edge boxes, 210 radars, 260 cameras)	$41,760.00
Staff training and rollout	$12,800.00
Dedicated engineering, model tuning and support	$31,650.00

Total for 12 months — $619,140.00

Paid monthly at $51,595.00 (first payment $51,595.00)

BUILD SHEET 1 OF 4

Your new platform · $233,200.00

FEATURE & PRICE	TECH YOU NEED	AI YOU NEED	GITHUB / OPEN SOURCE
Resident, family and care-plan engine $64,850.00 One record per resident: care plan, fall-risk score, consent forms, emergency contacts, family access, room/home assignment and device inventory.	Next.js + TypeScript, PostgreSQL (Supabase or AWS RDS), FHIR R4 data model, row-level security per facility.	Claude summarises care notes into a 3-line daily brief per resident; fall-risk score from history + gait trend.	medplum/medplum supabase/supabase hapifhir/hapi-fhir
Camera and sensor ingestion (edge NVR) $38,420.00 Connects PoE/Wi-Fi cameras, mmWave radars and wearables at every site into one secure edge box per building or home.	NVIDIA Jetson Orin Nano / Orin NX edge box, RTSP/ONVIF, go2rtc restreaming, MQTT bus, WireGuard tunnel to cloud.	Edge inference with TensorRT; only events and skeletons leave the building, never raw video by default.	blakeblackshear/frigate AlexxIT/go2rtc bluenviron/mediamtx eclipse/mosquitto
24/7 monitoring command center $31,275.00 Live wall for nurses and the central response desk: rooms, residents, alerts, escalation timers and acknowledgements.	React + WebRTC (sub-second video on demand), WebSockets, role-based access, audit log on every view.	Priority ranking of open alerts; VLM second-look caption on each event ("resident seated on floor beside bed").	livekit/livekit AlexxIT/go2rtc grafana/grafana
Caregiver mobile app (iOS and Android) $35,725.00 Staff get the alert, see the skeleton clip, tap Accept, and log the outcome. Hands-free voice acknowledge.	React Native (Expo), APNs/FCM critical alerts, offline queue, SSO (SAML/OIDC).	Voice acknowledge and dictated incident notes (Whisper), auto-filled into the report.	expo/expo novuhq/novu ggerganov/whisper.cpp
Family portal and app $18,640.00 Families see wellbeing (sleep, activity, check-ins), receive alerts they opt into and can video-call their loved one.	Same Expo codebase, separate role; consent-gated data views; Stripe billing for at-home plans.	Weekly AI wellbeing letter in plain language, written from activity data, never from video.	expo/expo stripe/stripe-node
EHR and nurse-call integration $24,310.00 Pushes incidents and vitals into PointClickCare / MatrixCare and triggers existing nurse-call systems.	HL7 v2 + FHIR interface engine, vendor APIs, dry-contact relay for legacy nurse call.	Maps free-text notes to coded fields (fall type, injury, location) for the EHR.	nextgenhealthcare/connect medplum/medplum
Moving your data over $19,980.00 Imports residents, rooms, contacts and incident history from spreadsheets or the current EHR.	ETL scripts, validation reports, rollback plan.	Claude-assisted field mapping and de-duplication with human sign-off.	dbt-labs/dbt-core

BUILD SHEET 2 OF 4

AI that does the watching · $182,470.00

FEATURE & PRICE	TECH YOU NEED	AI YOU NEED	GITHUB / OPEN SOURCE
Vision fall detection (privacy skeleton mode) $46,480.00 Detects falls, slides from bed/chair and residents on the floor in under 3 seconds, from pose keypoints instead of faces.	YOLO11-pose or RTMPose on the edge box, ByteTrack IDs, temporal classifier (ST-GCN) on 2-second keypoint windows.	Train/validate on UR Fall, Le2i and CAUCAFall + your own consented, labelled clips; target ≥ 95% recall, < 1 false alarm per room per week.	ultralytics/ultralytics open-mmlab/mmpose ifzhang/ByteTrack yysijie/st-gcn GajuuzZ/Human-Falling-Detect-Tracks
Bathroom and bedroom radar (no camera) $22,740.00 Fall, presence, breathing-rate and in-bathroom-too-long detection where cameras are never allowed.	60 GHz mmWave fall-detection radars (e.g. Seeed MR60FDA2, Aqara FP2, TI IWR6843) over Wi-Fi/Zigbee → MQTT.	Fusion rule engine: radar fall + no motion for 30 s + no wearable step = confirmed alert.	Seeed-Studio/Seeed_Arduino_mmWave home-assistant/core Koenkk/zigbee2mqtt
Wander, exit and bed-exit alerts $17,560.00 Virtual zones: memory-care doors, stairwells, bed edge at night, kitchen stove left on.	Polygon zones per camera, time-of-day schedules, door contacts and smart locks.	Tracker-based zone crossing with direction; night-time bed-exit predicted from restlessness 2-5 min early.	roboflow/supervision ifzhang/ByteTrack home-assistant/core
AI false-alarm verifier (second look) $21,390.00 Before anyone is paged, a vision-language model checks the clip: real fall, or a resident doing yoga on the rug?	Small VLM on the edge box (Moondream / Florence-2 / Qwen2.5-VL 3B), escalates to a cloud model when unsure.	Cuts false alarms 60-80% in pilots of this design; every suppressed alert is still logged for audit.	vikhyat/moondream QwenLM/Qwen2.5-VL microsoft/Florence-2-large (HF)
Acoustic distress detection $12,870.00 Hears 'help', crying out, a thud or breaking glass and links it to the room's camera/radar.	On-device audio classification; keyword spotting; no audio recorded or stored.	YAMNet sound classes + Whisper-tiny keyword spotting in English and Spanish.	google-ai-edge/mediapipe ggerganov/whisper.cpp
Daily-living baseline and decline alerts $19,950.00 Learns each resident's normal: sleep, bathroom visits, meals, steps, gait speed. Flags changes days before a crisis (UTI, dehydration, depression, rising fall risk).	Time-series store (TimescaleDB), nightly jobs, wearable data (Apple HealthKit, Health Connect).	Per-resident anomaly models (Isolation Forest / seasonal baselines) + Claude-written explanation for the nurse.	yzhao062/pyod timescale/timescaledb facebook/prophet
24/7 AI voice check-in concierge $26,820.00 Calls or speaks to the resident after an alert or on schedule: 'Are you alright, Margaret?' Escalates if no answer or distress.	Telephony (Twilio) + smart speaker in room, real-time voice agent pipeline.	Claude as the conversational brain, streaming STT/TTS; strict script guardrails, never gives medical advice.	pipecat-ai/pipecat livekit/agents twilio/twilio-node
AI incident reports and shift handover $14,660.00 Drafts the state-compliant incident report, family notification and end-of-shift summary from the event timeline.	Templating per state, e-signature, PDF export, stored in EHR.	Claude drafts from structured events + caregiver voice note; nurse reviews and signs.	anthropics/anthropic-sdk-typescript diegomura/react-pdf

BUILD SHEET 3 OF 4

Response, security and compliance · $68,060.00

FEATURE & PRICE	TECH YOU NEED	AI YOU NEED	GITHUB / OPEN SOURCE
Alert escalation engine $16,980.00 Nearest caregiver → charge nurse → on-call manager → family → 911, with timers and acknowledgements at every step.	Durable workflows, SMS/voice/push, on-call rotas, geolocated staff badges (optional BLE).	Picks the closest free caregiver from badge location and current workload.	temporalio/temporal novuhq/novu
Privacy, consent and HIPAA security $22,480.00 Consent per resident and roommate, camera privacy schedules, skeleton-only mode, encryption, full audit trail, BAA-ready hosting.	AWS (HIPAA-eligible services + BAA), KMS encryption, SSO/MFA, immutable audit log, SOC 2 readiness (Vanta/Drata).	On-device face blurring for any clip that is ever shared; no facial recognition anywhere.	open-policy-agent/opa ory/kratos
Operations and compliance analytics $13,240.00 Falls per 1,000 resident-days, response times, repeat fallers, staffing vs incidents, state-survey-ready exports.	Metabase / Superset dashboards on a read replica.	Ask-in-English analytics: 'Which wing had the slowest night responses last month?'	metabase/metabase apache/superset
Website, online enrollment and at-home ordering $15,360.00 Brand site, facility demo booking, family self-enrollment for at-home plans, hardware checkout.	Next.js, Stripe, Calendly/Cal.com, CMS.	AI chat on the site that answers plan, privacy and pricing questions and books demos.	calcom/cal.com vercel/next.js

BUILD SHEET 4 OF 4

Ongoing support, hosting and usage (12 months) · $135,410.00

FEATURE & PRICE	TECH YOU NEED	AI YOU NEED	GITHUB / OPEN SOURCE
Cloud hosting, security and backups ($2,140/mo) $25,680.00 HIPAA-eligible AWS, video-on-demand relay, encrypted backups, uptime monitoring.	AWS, Cloudflare, Grafana/Prometheus.	—	`prometheus/prometheus`
AI usage, voice and messaging ($1,960/mo) $23,520.00 LLM calls, voice minutes, SMS and push volume across all sites.	Anthropic API, Twilio, FCM/APNs.	Claude (Haiku for routine, Sonnet/Opus for reports) with prompt caching.	`service line`
Edge hardware lease (34 edge boxes, 210 radars, 260 cameras) $41,760.00 Leased, insured and replaced on failure. At-home kits ship pre-paired.	Jetson Orin, PoE cameras, mmWave radars, SOS wearables.	—	`service line`
Staff training and rollout $12,800.00 On-site go-live at each community, train-the-trainer, family onboarding webinars.	LMS videos, printed quick cards.	AI training assistant inside the caregiver app.	`service line`
Dedicated engineering, model tuning and support $31,650.00 Named engineer, 24/7 P1 support, quarterly fall-model retraining on your labelled events.	SLA 99.9%, 15-minute P1 response.	Continuous evaluation: recall, false alarms per room-week, time-to-alert.	`service line`

RUNNING VIGIL AS AN ENTERPRISE

How the business works.

Who pays

- Care communities (assisted living, memory care, skilled nursing): per-bed SaaS at $45-79 per bed per month on 3-year terms, plus a one-time build and integration fee for large operators (this proposal).
- Home-care agencies: white-label VIGIL for their clients; remote eyes between visits.
- Families direct: a $699 home kit (edge hub, 2 radars, 1 camera, SOS wearable) plus $129-249 per month (self-monitored up to 24/7 concierge response).
- Payers: Medicaid waiver programs and some Medicare Advantage plans cover personal emergency response; long-term-care insurers value documented fall prevention.

Unit economics

	Facility bed	At-home plan
Price / month	$59	$189
Cost to serve / month	$14	$41
Gross margin	76%	78%
Example: 540 beds + 250 homes	$31,860 MRR	$47,250 MRR
Combined annual recurring revenue		**$949,320 ARR**

Cost to serve includes cloud, AI usage, hardware amortisation and a share of the 24/7 response desk. The home kit costs about $310 to build.

What must be right before launch

- Consent and state law. Several states regulate electronic monitoring in resident rooms (for example Texas, Illinois, Oklahoma, Washington, Virginia): written consent from the resident or legal representative and any roommate, posted notice, and rules on access to footage.
- HIPAA. VIGIL acts as a business associate: BAAs with every customer and sub-processor, a documented risk assessment, breach procedures.
- Claims. Market as safety and wellness monitoring that alerts people, not as a diagnostic medical device. Have a regulatory attorney review all marketing claims.
- Measured accuracy. Publish recall, false alarms per room-week and time-to-alert from your own pilots.
- Insurance: technology E&O, cyber liability and general liability before the first install.

The first 120 days

Days 0-30	Entity, insurance, BAA templates. Edge box running Frigate + YOLO11-pose + radar fusion. Recruit one friendly community for a free 10-suite pilot.
Days 30-60	Pilot live with consent packs. Label every event; tune until false alarms < 1 per room per week. Ship the caregiver app.
Days 60-90	Turn pilot numbers into a case study (time to bedside, falls with injury, night-round hours saved). Launch the home kit to 25 families.
Days 90-120	Sell to regional operators (5-20 communities) through state assisted-living associations; partner with two home-care agencies.

Core team

Founder (sales, brand, operator relationships) · full-stack engineer · ML / edge engineer · installer and field technician · clinical advisor (RN, director of nursing) · part-time compliance lead. The 24/7 response desk can start with a UL-listed monitoring partner.

NEXT STEPS

Let's begin with ten suites.

We start with a 60-day paid pilot in ten suites at one community, priced at cost and credited in full against this agreement. If VIGIL does not cut time-to-bedside in half against your current baseline, you walk away.

- Week 1: site walk, consent packs, network survey.
- Week 2: install edge box, radars and cameras; train night staff.
- Weeks 3-8: live pilot with weekly scorecards.
- Week 9: go / no-go and portfolio rollout schedule.

Colin Cook · Founder, VIGIL	Client signature · Title · Date

Live demo console: bm6k4rc72b-droid.github.io/coach-colin-ai-command-center/vigil/
Resident names, facilities and outcomes in the demo are fictional. Statistics marked CDC are from the U.S. Centers for Disease Control and Prevention. Accuracy and response figures are design targets to be validated in the pilot.